# Array and Object

### Answer to Array and Object Assignment

### 1. Shopping Cart Operations

```
const shoppingCart = ['Milk', 'Coffee', 'Tea', 'Honey'];

// Add 'Meat' at the beginning if not already present
if (!shoppingCart.includes('Meat')) {
    shoppingCart.unshift('Meat');
}

// Add 'Sugar' at the end if not already present
if (!shoppingCart.includes('Sugar')) {
    shoppingCart.push('Sugar');
}

// Remove 'Honey' if allergic
const allergic = true;
if (allergic) {
    const honeyIndex = shoppingCart.indexOf('Honey');
    if (honeyIndex !== -1) {
        shoppingCart.splice(honeyIndex, 1);
    }
}

// Modify 'Tea' to 'Green Tea'
const teaIndex = shoppingCart.indexOf('Tea');
if (teaIndex !== -1) {
    shoppingCart[teaIndex] = 'Green Tea';
}

console.log(shoppingCart);
```

### 2. Student Ages Operations

```
const ages = [19, 22, 19, 24, 20, 25, 26, 24, 25, 24];

// Sort the array
ages.sort((a, b) => a - b);
console.log("Sorted ages:", ages);
```

```javascript
// Find min and max
const minAge = ages[0];
const maxAge = ages[ages.length - 1];
console.log("Min Age:", minAge, "Max Age:", maxAge);

// Find median
const medianAge = ages.length % 2 === 0 ?
    (ages[ages.length / 2 - 1] + ages[ages.length / 2]) / 2 :
    ages[Math.floor(ages.length / 2)];
console.log("Median Age:", medianAge);

// Find average
const averageAge = ages.reduce((sum, age) => sum + age, 0) / ages.length;
console.log("Average Age:", averageAge);

// Find range
const ageRange = maxAge - minAge;
console.log("Age Range:", ageRange);

// Compare (min - avg) and (max - avg)
console.log("Min - Avg:", Math.abs(minAge - averageAge));
console.log("Max - Avg:", Math.abs(maxAge - averageAge));
```

### 3. Object Extensibility and Sealing

```javascript
const student = { name: "John", age: 22 };
Object.preventExtensions(student);
const extensibleStatus = Object.isExtensible(student);
console.log("Extensible Status:", extensibleStatus);

const teacher = { subject: "Math" };
Object.seal(teacher);
const sealedStatus = Object.isSealed(teacher);
console.log("Sealed Status:", sealedStatus);
```

### 4. Student Management System

```javascript
let students = [];

function addStudent(id, firstName, lastName, age, grade) {
    students.push({ id, firstName, lastName, age, grade });
}

function updateStudent(id, newInfo) {
```

```
    let student = students.find(s => s.id === id);
    if (student) {
        Object.assign(student, newInfo);
    }
}

function deleteStudent(id) {
    students = students.filter(s => s.id !== id);
}

function listStudents() {
    console.log(students);
}

function findStudentsByGrade(grade) {
    return students.filter(s => s.grade === grade);
}

function calculateAverageAge() {
    return students.reduce((sum, s) => sum + s.age, 0) / students.length;
}
```

### 5. Display Student Info

```
function displayStudentInfo(student) {
    for (let key in student) {
        console.log(`${key}: ${student[key]}`);
    }
}
```